
Retention You Can Predict, Scope and Delete: a (μ, γ, T) Law and a Structural Deletion Guarantee for a Physics-Structured Memory Store

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 What a memory system forgets is set by policy: a decay factor, an expiry timestamp,
2 a learned forget gate. Such a policy states when an entry expires, not how fast
3 the stored value degrades, what moving the dial buys, or what remains once it is
4 removed. We answer all three on a physics-structured memory store — a damped
5 symplectic step of a learned Hamiltonian — whose retention policy is a budget
6 in three numbers: the stored direction’s spectral mass μ , the damping γ and the
7 temperature T . The dial has a computable optimum: retention half-life is non-
8 monotone in friction, one V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$ ($\propto \mu^{-2}$ overdamped,
9 a mass-independent floor underdamped), spanning ≈ 11 orders of magnitude in
10 μ^2 , holding on emergent (learned-MLP) as well as designed units, 3/3 seeds (dim
11 4, hidden 64, laptop-CPU), and reproduced by direct nonlinear rollout with the
12 argmin agreeing to 0.35%. Retention is scopeable: at $T > 0$ friction preserves and
13 temperature erases, so a localized friction hole is a memory vault, $107.77 \pm 4.78 \times$
14 designed and $106.1 \pm 5.0 \times$ law-referenced on emergent (3 seeds each), though the
15 field-versus-scalar contrast naming its mechanism is designed-only. Removal is
16 structural: under canonical placement — Brelloch & Golovin’s (FOCS’07) — store-
17 level deletion is exact, the post-deletion store byte-identical at every load measured
18 to the store that never held the item, under three stated conditions (budget $\geq$
19 n_cells, leak = 0, priority/attribute eviction; recency eviction excluded), on a
20 designed store of dim 3, capacity 8–64, no learning. We state what decay does
21 not buy: against an exact adversary a boolean TTL flag in the same store is as
22 detectable as physical decay, and the store stops answering before it stops leaking.

23 1 Introduction

24 Every deployed memory store already runs a retention policy: a decay factor, an invalidation
25 timestamp, a learned forget gate, a DELETE call. Forgetting, however, is the failure mode benchmarks
26 do not measure (Yang, 2026; Uddin et al., 2026), and a policy is not a law. One cannot read off
27 a trained LSTM the half-life of a stored value, the knob that sets it, or what remains once the
28 value is removed. We study a memory whose forgetting is a dynamical property of the store, not
29 a bookkeeping rule over it, so that three operational questions have computable answers: *what is*
30 *the retention policy, and where is its optimum* (§3.1); *can retention be scoped to one item* (§3.2); *is*
31 *deletion real, and what still leaks* (§3.3).

32 Physics enters only as the derivation apparatus. The store’s state (q, p) advances by a damped
33 symplectic step of a learnable Hamiltonian $H = T(p) + V_\theta(q)$: the damping $p \mapsto (1 - \gamma)p$
34 supplies forgetting, Langevin noise supplies temperature T , and items are optionally written as *atoms*:
35 localized wells superposed into one energy function a dynamical relaxation reads. Our reference
36 unit is the CLU (Causal Learning Unit), introduced as CHLU in Jawahar & Pierini (2026). Its

core is exactly solvable: diagonalised in the mass-whitened Hessian, one damped step acts on each normal mode as a single 2×2 matrix fixed by $(\varepsilon\mu, \gamma)$, whose spectral radius is that mode’s retention (Anonymous, 2026, a theory note on which nothing here depends).

Contributions. (1) *A retention dial with a computable optimum* (§3.1–§3.2): retention is non-monotone in the dial and the optimum is predicted, not tuned — one V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$, so latch, overdamped register and underdamped working memory are three regimes of one curve, evaluated at two values of μ , not two laws, across ≈ 11 decades in μ^2 , on emergent as well as designed units, on two instruments. (2) *Scoped retention* (§3.2): at $T > 0$ friction preserves and temperature erases, so a local change to the dial confines one item’s coordinate, and a friction hole is a vault whose two constituent laws hold on both arms. (3) *A structural deletion guarantee* (§3.3), a composition and not a new deletion algorithm: the placement rule and its delete-time repair are Blleloch & Golovin’s (FOCS’07); new here is only the composition — a lattice packing certificate, decaying content whose decay provably commutes with deletion, one canonical energy function read by a dynamical relaxation — which makes the store’s state a function of its live set alone, plus the trilemma that prices lifetimes and an honest leakage statement. (4) *A three-state memory lifecycle* (PROTECTED $\rightleftharpoons$ ACTIVE $\rightarrow$ TRASH, 7/7 legs, kill-conditions committed before the verbs, one leg unexercised) — the consolidation and stale-entry face of the same store — shipped as a mechanics demonstration on a toy synthetic build: no value or benchmark number is claimed, and none was run.

What class of claim this is. Our evidence is a small designed store measured at laptop scale: not a deployed agent memory, not an LLM system, not a benchmark result. We report a mechanism with measured laws, not a system result.

Nomenclature, because “mass” names two opposed quantities. The inertial mass M_i is the learned diagonal of the kinetic term (larger $\Rightarrow$ slower); the spectral mass is $\mu_k^2 = \lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$ (larger $\Rightarrow$ shorter memory). Retention statements use μ ; a stored value lives on a near-flat coset direction; ε is the integrator step, $= 0.05$, never a tilt. Half-lives mean nothing without a read tolerance Δ and the thermal persistence length ℓ_θ , so we never quote an $n_{1/2}$ without its Δ and ℓ_θ/Δ , and never read a designed-versus-learned distinction off raw exponents. Results on designed testbeds are labelled verification of exactness; on learned or emergent potentials, evidence. Every $T > 0$ result requires the fluctuation-dissipation-consistent noise scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$ and a Newtonian kinetic mode. These laws govern the store; end-to-end performance additionally depends on the encoder, measured separately, φ -bytes ledgered.

2 Related work

Retention policies in deployed memory systems. Agent memories are external stores a model writes to, retrieves from and edits (MemGPT, Packer et al., 2023; Mem0, Chhikara et al., 2025), or bounded compressive state inside the network (Infini-attention, Munkhdalai et al., 2024). Their retention policies are scores and gates: a 0.995 recency decay factor in a ranking heuristic (Generative Agents, Park et al., 2023), an Ebbinghaus decay of memory strength (MemoryBank, Zhong et al., 2024), a learned per-memory expiration span (Expire-Span, Sukhbaatar et al., 2021), and a learned forget gate α_t in $\mathcal{M}_t = (1 - \alpha_t)\mathcal{M}_{t-1} + S_t$ (Titans, Behrouz et al., 2025) — the nearest published neighbour to the dial of §3.1. None states a half-life, a read tolerance or a temperature, and none prices what is lost. We benchmark against none of them; the contrast is structural, our (μ, γ, T) budget occupying the same slot with those three quantities attached.

Deletion, deployed and formal. Zep invalidates a contradicted edge with a timestamp and preserves the record (Rasmussen et al., 2025); Mem0’s DELETE removes a row. Byte-identity is the opposite design point, and it matters because soft-deleted vectors stay reconstructible from raw index files in graph ANN databases (Chakrabortii et al., 2026). A 2026 evaluation wave motivates the framing: production failures are predominantly forgetting failures while benchmarks measure recall (Yang, 2026), failures to forget outdated memory drive a large share of sampled recommendation errors (Uddin et al., 2026), deleted facts stay recoverable through retained correlated modalities (Wang & Zhang, 2026), and agentic unlearning must handle reactivation out of persistent memory (Wang et al., 2026) — the last two being why §3.3’s encoder-excluded scope is a scope, not a hedge: the store deleting an item is not the system forgetting it. Formally, *certified* removal is Guo et al.’s (2020) §2 Eq. (1), an ε condition with an unnumbered (ε, δ) relaxation immediately after; exact methods delete by isolation, with cost and capacity formalisms in place (Bourtole et al., 2021; Ginart et al., 2019; Sekhari et al., 2021). We make none of those claims: ours is a store-level bit-exactness statement with the encoder excluded, coining no benchmark and no cost claim. Order-independent placement is not ours either — Snyder (1977) through Naor & Teague (2001) to Blleloch & Golovin (2007), whose table we use; and Blleloch, Golovin & Vassilevska (2008) had already taken unique representation into computational geometry, so canonical placement in a geometric setting is not virgin ground (Appendix D).

Forgetting laws, learned and physical. Learned recurrent comparators (LSTM, Hochreiter & Schmidhuber, 1997; LEM, Rusch et al., 2022) lack a stateable forgetting law. In physics, dissipation turns type-A Nambu–Goldstone modes from propagating to diffusive (Minami & Hidaka, 2018) — exactly our coset — and Mo (2026) proves that an exactly G -equivariant flow has *at least* $\dim(G/\mathcal{H})$ zero Lyapunov exponents along the orbit: the kinematics of protection, to which our budget is the constitutive complement, since the gap is μ^2 .

3 Results

3.1 The retention dial has a computable optimum

If damping is the retention dial, where should it be set? Not by tuning, and not at the low end: retention half-life is non-monotone in friction, and the turning point is fixed by the stored direction’s spectral mass, so the optimum is predicted.

The step is conformally symplectic, $\det J = (1 - \gamma)^d$, and $(1 - \gamma\phi(q'))^d$ when position-gated; checkpoints (dim 4, hidden 64, $\varepsilon = 0.05$, 150 epochs) are *designed* ($SO(2)$ -invariant V_θ , 5 seeds) or

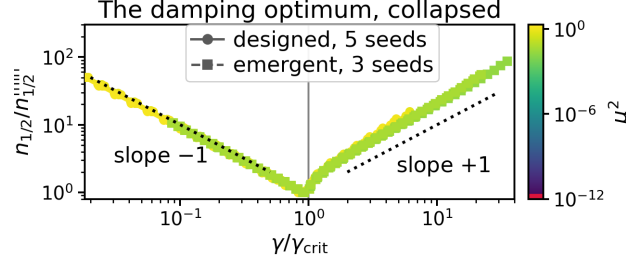


Figure 1: *The retention dial and its optimum, collapsed (headline)*. Eight measured curves collapse onto one: five designed radial modes (circles, verification, $\mu_{\text{rad}}^2 = 0.670\text{--}1.348$) and three emergent learned-MLP coset modes (squares, evidence, $\mu_{\text{soft}}^2 = 2.0\text{--}5.4 \times 10^{-2}$, 3/3 seeds), against the ∓ 1 asymptotes; the designed flat coset is the $\mu \rightarrow 0$ corner of the same curve. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; langevin_noise="fdt", Newtonian kinetic mode.*)

97 *emergent* (MLP V_θ on identical symmetric data, 3 seeds). On a trained designed- $SO(2)$ vacuum the
 98 massive radial mode's $n_{1/2}(\gamma)$ is a V minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu_{\text{rad}}$, the overdamped branch following
 99 the μ^{-2} law up to $\varepsilon\mu \approx \gamma/2$ and then saturating at a mass-independent floor (log-slopes -1.006
 100 and $+1.23\text{--}1.27$; 5 seeds, verification, Appendix A). The flat coset (Hessian $\mu^2 \approx 10^{-15}$) is the
 101 *same curve* at $\mu \rightarrow 0$, where $\gamma_{\text{crit}} \rightarrow 0$: latch, overdamped register and underdamped working
 102 memory are three regimes of one curve, evaluated at two values of μ — not two laws. It generalizes
 103 to emergent units (evidence): on three emergent MLP checkpoints trained on the same symmetric
 104 data the coset-tracked $T = 0$ half-life traces the same V-curve — argmin $0.902 \pm 0.003 \times \gamma_{\text{crit}}$,
 105 log-slopes $-1.0020 \pm 0.0003 / +1.116 \pm 0.011$, 3/3 seeds. Across both families the curve spans
 106 $\mu^2 \in [1.7 \times 10^{-12}, 7 \times 10^{-2}]$, that low endpoint being the ring-profile probe's resolution on a
 107 checkpoint whose Hessian μ^2 is machine zero rather than a spectral mass — so eleven orders is one
 108 curve on one instrument, though the $\mu \rightarrow 0$ corner has its own model-side bound of $n_{1/2} \geq 9.5 \times 10^{15}$
 109 steps by direct $T = 0$ rollout (Appendix B).

110 *Two instruments, one law (evidence)*. Every point above is a one-step Jacobian, so we re-measured
 111 the curve by direct nonlinear rollout, reading $n_{1/2}$ from the decay envelope's rate (3 emergent
 112 seeds, $\delta = 0.05\text{--}0.5$ rad). The shape reproduces: argmin 0.9001 ± 0.0052 against the Jacobian's
 113 0.9032 ± 0.0027 in units of γ_{crit} — a 0.35% difference — log-slopes $-0.983 \pm 0.013 / +1.117 \pm 0.010$
 114 on identical fit windows, and decay *rates* agreeing to 0.995 ± 0.003 in the asymptotic window. The
 115 instruments differ by a level, not a law: a first-crossing threshold reads short by a per-checkpoint
 116 constant $(1.33/1.80/1.82 \times)$ flat in γ , with coefficient of variation 2.2–3.7%. *What would have falsified the*
 117 *law*: a stored direction that mixes normal modes, or is anharmonic at the write amplitude, has no single-mode reduction, so the curves would
 118 neither collapse nor put their minimum at $2\varepsilon\mu$; neither happens at ten times the write amplitude, where the argmin moves by at most 1.4% (to
 119 $0.897\text{--}0.917 \gamma_{\text{crit}}$) and the slopes by at most 1.9% (Appendix B).

120 3.2 Scoping retention to one item: friction preserves, temperature erases

121 Scoping a policy to one entry means changing the dial locally, and the sign of the dial's effect on
 122 a stored value is not the expected one. At $T = 0$ a designed coset latches forever at any friction
 123 (drift $\leq 4.9 \times 10^{-12}$ rad over 200k steps, all $\gamma \in [0.002, 0.5]$, 5 seeds, verification): friction has no
 124 contractive action on a flat direction. At $T > 0$ the coset decays diffusively and friction slows it down
 125 — $D_\theta = \varepsilon T(2 - \gamma)/(2F^2\gamma)$ is verified to 1.0068 ± 0.0219 over 25 (γ, T) cells, so $n_{1/2} \propto \gamma^{+1}$ and the
 126 sign flip $\partial n_{1/2}/\partial \gamma > 0$ holds in 10/10 seed $\times$ temperature conditions, $\gamma : 0.05 \rightarrow 0.2$ *lengthening*
 127 the half-life $3.77 \pm 0.23 \times$ at $\Delta = 0.5$ rad, $\ell_\theta/\Delta < 0.05$; only temperature has $\partial n_{1/2}/\partial T < 0$. *Fine*
 128 *print, beside the claim*: every $T > 0$ number requires langevin_noise="fdt" and a Newtonian kinetic mode; under the reference default
 129 these laws fail, and in the relativistic mode no noise scale targets Gibbs.

130 *Hence a memory vault*. The shipped position-gated friction field is *absorb-only*, so a hole is a brake
 131 and a refrigerator ($T_{\text{local}} = 1.26 \times 10^{-4}$ against 10^{-3} outside, where a coupled bath predicts none),
 132 and the vault factor is $(\gamma_{\text{eff}}/\gamma)^2 = 107.77 \pm 4.78 \times$ (D_θ -estimator, pre-registered, 3 designed seeds,
 133 $\Delta = 0.5$ rad) against $13.28 \pm 0.12 \times$ for a scalar friction of equal γ_{eff} , the ratio 8.11 ± 0.37 being
 134 the refrigerator. At $T = 0$ it erases nothing. Both laws transfer to an emergent register (evidence,
 135 3 emergent seeds at $T = 4 \times 10^{-3}$ and 8×10^{-3} , both above T^*): the refrigerator law holds at
 136 0.9998 ± 0.0019 of prediction (the coupled bath rejected again at 0.2235) and the $\hat{D}_\theta \propto \gamma_{\text{eff}}^{-2}$ law
 137 at 1.016–1.103, a law-referenced vault of $106.1 \pm 5.0 \times$ beside the designed $107.77 \pm 4.78 \times$. One
 138 result has no designed analogue, and it is the one that makes scoping operational — *the hole confines*:
 139 the fraction of states outside the register ($|\theta| > 1$ rad) falls from 5.5/43.0/2.4% with no hole to

0.0000 inside it, 3/3 seeds, while a scalar friction of the same γ_{eff} still hops (0.73/10.2/0.26%). *The contrast number stays designed-only:* on emergent the field/scalar ratio measures 23.39 ± 10.06 because the *control* arm delocalises there ($\bar{D}/D_{\text{flat}} = 1.18\text{--}5.23$) while the field arm sits on the law, so 8.11 ± 0.37 is a designed quantity and the direction, not the number, transfers. We quote no emergent σ_θ ratio: above T^* that diagnostic is non-stationary here, its control returning 0.459 ± 0.118 where it must return 1.000 (Appendix C).

The designed-symmetry precondition, the boundary on everything above (evidence; 3 emergent seeds and a matched designed control). The exact $T = 0$ latch is a *designed* object: on an MLP potential trained on identical symmetric data the softest direction is a mid-spectrum massive mode ($1 - |\lambda_{\text{coset}}| \approx 10^{-3}$ against the designed $\leq 1.1 \times 10^{-15}$), a written value relaxes completely, and capacity is $\approx 1\text{--}1.6$ bits. An emergent unit of this class has no continuous coset register, and continuum flat directions must be designed in. What survives is exactly the headline — the V-curve on 3/3 seeds and the sign flip above a measured $T^* \approx 3 \times 10^{-3}$. The same boundary bounds an extrapolation we do not make, and Appendix E records it.

3.3 Deletion: a structural guarantee, priced lifetimes, and what still leaks

(Designed atom store, dim 3, no learning anywhere $\Rightarrow$ verification.) Deletion in a deployed store is best-effort: a row dropped, an edge timestamped, a tombstone written. We ask instead whether the store’s state can be made a function of its live set alone. Stated once: external benchmarks won on their own headline metric = ZERO. A flat table deletes exactly by construction — the trivial substitute; at issue is a store whose items are *superposed into one energy function*, where placement would ordinarily depend on which others are present. Placement here is canonical, a function of the live records and the geometry alone, so removal reproduces the store holding the rest bit for bit. Measured: “with the waitlist, store-level deletion is exact at every load measured, $0.29 \times \text{--} 1.71 \times$ lattice capacity — AUC 0.5000 ± 0.0000 on all six statistics, byte-equal 1.0000 — under three stated conditions (budget $\geq n_{\text{cells}}$, leak = 0, priority/attribute-based eviction; recency-based eviction remains intrinsically history-dependent and is excluded).” The allocator it replaces leaks membership after eviction (AUC 0.99985 at full load), and the packing price of exactness is negative: canonical placement admits strictly more keys than refuse-and-relocate. This is a store-level guarantee only — the frozen encoder and any residue of past writes in a learned landscape are separate channels, measured separately, and we claim no certified (ϵ, δ) unlearning; without the waitlist, exactness fails at overflow (Appendix D).

Attribution, in the same breath as the theorems. Our placement rule is the Blerloch–Golovin (FOCS’07) stable-matching table, our Theorems 1–2 follow from theirs, and the fix-up cascade is theirs. The composition’s price is negative where the general case can cost an exponential slowdown (Buchbinder & Petrank, 2003). That separation is between the *weak* and *strong* notions for comparison-based structures, whereas ours is against a history-dependent stochastic relocation rule, so the two statements are adjacent rather than identical.

Lifetimes are priced by a trilemma — exact value fidelity, amplitude-independent address hold, amplitude-independent read latency: pick two. The corner to lead with: *dropping amplitude-independent latency is the compute-adaptive-read dial — a faded memory costs more integration steps to read, a physical, measurable statement a timestamped row cannot make.*

What decay does not buy. The tighter laundering control is the policy already deployed elsewhere, a boolean TTL flag *inside the same store*, and it fires: against an exact adversary the TTL flag stays within 0.017 AUC of physical decay (0.983 versus 1.000), the two separating only against a resolution-limited adversary (0.559 versus 0.996 at $\sigma_{\text{obs}} = 0.1$), whose resolution is our own modelling choice. *The store stops answering before it stops leaking:* at the last amplitude before self-eviction ($A = 0.051$) retention is 0.832 — placement-dependent, quoted for the controller-placed disk — while a query-level membership adversary sits at AUC 0.983 and a white-box one at 1.000; the curves never cross. Against an *exact* adversary decay reduces the effect size, not the AUC, so we claim no reduction in distinguishability per se. What physical decay buys over a TTL flag is retrieval geometry, not privacy, and that differentiator needs no adversary model: *decay contracts the addressing tolerance of an item, $R_{50} 1.146 \rightarrow 0.752$ ($1.52 \times$), where a TTL dict’s lookup radius is a constant hard step at 0.75–0.77, independent of age.*

4 Limitations and future work

Limitations. (i) One architecture, one scale: dim 4, hidden 64, laptop-CPU, 5 designed / 3 emergent seeds; store at dim 3, no learning; lifecycle on a toy synthetic build. (ii) The designed-symmetry precondition (§3.2). (iii) Raw T/γ -exponents do not discriminate designed from emergent; the $\text{fdt} + \text{Newtonian}$ scope is mandatory at every $T > 0$. (iv) The V-curve’s shape is instrument-independent, its absolute step counts are not (a first-crossing threshold reads $1.33\text{--}1.82 \times$ short of the Jacobian, a constant per-checkpoint offset), and the emergent collapse spans only $2.7 \times$ in μ^2 across three seeds, so the eleven decades are one decade of emergent variation attached to the designed $\mu \rightarrow 0$ corner. (v) Deletion is store-level only, under set-function eviction with recency eviction excluded, and no deletion-cost claim is made ($O(n)$ rebuild per operation; unrun). (vi) No task-level payoff is claimed. *Scale is a scope choice:* this short is sealed to laptop-scale evidence; no larger-scale result is quoted or should be inferred. *Named next:* a wider emergent collapse in μ^2 ; the vault’s contrast arm re-measured where both arms stay bounded; amortized per-delete cost against the flat-datastore substitute; deletion at 10^3 -item stores; and a localized *temperature* field.

- Andersson, A., & Ottmann, T. (1995). New tight bounds uniquely represented dictionaries. *SIAM Journal on Computing* 24(5):1091–1103. 251
- Anonymous (2026). *The theory note* (exactly-solvable damp symplectic retention theory). Anonymized for double-blind review. 254
- Behrouz, A., Zhong, P., & Mirrokni, V. (2025). Titans: Learn to memorize at test time. *NeurIPS 2025*. arXiv:2501.00663. 256
- Blleloch, G. E., & Golovin, D. (2007). Strongly independent hashing with applications. *48th IEEE Symposium on Foundations of Computer Science (FOCS)*, 272–282. doi:10.1109/FOCS.2007.36. 260
- Blleloch, G. E., Golovin, D., & Vassilevska, V. (2008). Unique represented data structures for computational geometry. *Scandinavian Workshop on Algorithm Theory (SWAT)*, LNCS 5124:17–28. doi:10.1007/978-3-540-69903-3_4. 264
- Bourtole, L., Chandrasekaran, V., Choquette-Choo, C. A., H., Travers, A., Zhang, B., Lie, D., & Papernot, N. (2021). chine unlearning. *42nd IEEE Symposium on Security and Privacy*. arXiv:1912.03817. 268
- Buchbinder, N., & Petrank, E. (2003). Lower and upper bounds obtaining history independence. *CRYPTO 2003*, LNCS 2729:427–462. Journal version: *Information and Computation* 204(2):237–337 (2006). 272
- Chakrabortii, C., García Alvarado, J., Abdolozifova, S., & Dwivedi, S. (2026). Ghost vectors: Soft-deleted beddings remain reconstructible in HNSW vector databases. arXiv:2606.18497 (preprint). 276
- Chhikara, P., Khant, D., Aryan, S., Singh, T., & Yadav, D. (2023). Mem0: Building production-ready AI agents with scalable long-term memory. arXiv:2504.19413 (preprint). 279
- Ginart, A., Guan, M. Y., Valiant, G., & Zou, J. (2019). Maki AI forget you: Data deletion in machine learning. *NeurIPS 2019*. arXiv:1907.05012. 282
- Guo, C., Goldstein, T., Hannun, A., & van der Maaten, L. (2023). Certified data removal from machine learning models. *ICML 2023*. PMLR 119:3832–3842. arXiv:1911.03030. 285
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation* 9(8):1735–1787. doi:10.1162/neco.1997.9.8.1735. 288
- Jawahar, P., & Pierini, M. (2026). *Causal Hamiltonian Learning Units (CHLU)*. [Reference redacted for double-blind review.] 290
- Micciancio, D. (1997). Oblivious data structures: Application to cryptography. *29th ACM Symposium on Theory of Computing (STOC)*, 456–464. doi:10.1145/258533.258638. 293
- Minami, Y., & Hidaka, Y. (2018). Spontaneous symmetry breaking and Nambu-Goldstone modes in dissipative systems. *Physical Review E* 97(1):012130. doi:10.1103/PhysRevE.97.012130. 297
- Mo, H. H. (2026). Symmetry-protected Lyapunov neutral modes in equivariant recurrent networks. arXiv:2605.03338 (preprint; single author). 298
- Munkhdalai, T., Faruqi, M., & Gopal, S. (2024). Leave no context behind: Efficient infinite context transformers with Infinite attention. arXiv:2404.07143 (preprint). 299
- Naor, M., & Teague, V. (2001). Anti-persistence: History independent data structures. *33rd ACM Symposium on Theory of Computing (STOC)*, 492–501. doi:10.1145/380752.380844. 300
- Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S. G., Stoica, I., & Gonzalez, J. E. (2023). MemGPT: Towards LLMs as operating systems. arXiv:2310.08560 (preprint). 301
- Park, J. S., O’Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulations of human behavior. *UIST 2023*. arXiv:2304.03442. 302
- Rasmussen, P., Paliychuk, P., Beauvais, T., Ryan, J., & Chalef, D. (2025). Zep: A temporal knowledge graph architecture for agent memory. arXiv:2501.13956 (preprint). 303
- Rusch, T. K., Mishra, S., Erichson, N. B., & Mahoney, M. W. (2022). Long expressive memory for sequence modeling. *ICLR 2022*. arXiv:2110.04744. 304
- Sekhari, A., Acharya, J., Kamath, G., & Suresh, A. T. (2021). Remember what you want to forget: Algorithms for machine unlearning. *NeurIPS 2021*, 18075–18086. arXiv:2103.03279. 305
- Snyder, L. (1977). On uniquely representable data structures. *18th IEEE Symposium on Foundations of Computer Science (FOCS)*, 142–146. 306
- Sukhbaatar, S., Ju, D., Poff, S., Roller, S., Szlam, A., Weston, J., & Fan, A. (2021). Not all memories are created equal: Learning to forget by expiring. *ICML 2021*. arXiv:2105.06548. 307
- Uddin, M. N., Shubham, K., Blanco, E., Baral, C., & Wang, G. (2026). From recall to forgetting: Benchmarking long-term memory for personalized agents. arXiv:2604.20006 (preprint). 308
- Wang, B., Wang, F., Wang, P., Cong, J., Yu, Y., Yin, Y., Han, Z., & Wei, B. (2026). Agentic unlearning: When LLM agent meets machine unlearning. arXiv:2602.17692 (preprint). 309
- Wang, K., & Zhang, C. (2026). MemLeak: Diagnosing information leaks in multimodal agent memory. arXiv:2606.29788 (preprint). 310
- Yang, D. (2026). Control-plane placement shapes forgetting: An architectural study of agent memory across thirteen system configurations. arXiv:2606.15903 (preprint). 311
- Zhong, W., Guo, L., Gao, Q., Ye, H., & Wang, Y. (2024). Memory-Bank: Enhancing large language models with long-term memory. *AAAI 2024*. arXiv:2305.10250. 312

294 A The (μ, γ, T) budget: the diffusion law and the sign flip

295 *Designed $SO(2)$ testbed $\Rightarrow$ verification. All $T > 0$ cells require `langevin_noise="fdt"` and a Newtonian kinetic mode; the reference*
 296 *default is `legacy`, under which none of these laws hold. Every $n_{1/2}$ is quoted with its read tolerance $\Delta = 0.5$ rad and its ℓ_θ/Δ .*

297 *Diffusion law, 25 cells ($5\gamma \times 5T$, seed 44): $\hat{D}_\theta/D_\theta^{\text{pred}} = 1.0068 \pm 0.0219$ (min 0.9644, max 1.0484); $d \log D/d \log T =$
 298 $+0.998, +0.997, +0.992, +1.010, +1.006$; $d \log \hat{D}/d \log \gamma = -1.014, -0.980, -0.999, -1.001, -0.982$ after dividing the
 299 discrete $(2 - \gamma)$ factor. *Sign flip:* $d \log n_{1/2}/d \log \gamma = +0.9552 \pm 0.0422$ at $T = 10^{-3}$, 5 seeds, positive in 10/10 conditions; effect size
 300 $\gamma : 0.05 \rightarrow 0.2$ is $3.77 \pm 0.23 \times$ (per-seed 3.55, 3.64, 3.61, 3.89, 4.15); $d \log n_{1/2}/d \log T = -0.956$ to -0.979 deep-diffusive.
 301 *Latch at $T = 0$, designed only:* $|\lambda_{\text{flat}} - 1| \leq 1.7 \times 10^{-14}$ over 22 γ values on 5 seeds, drift $\leq 4.9 \times 10^{-12}$ rad per 200k steps in
 302 30/30 cells, against a $\gamma = 0$ control drifting 142.7 rad per 20k. *Designed V-curve:* argmin 0.076–0.096 against a predicted 0.082–0.116
 303 (per-seed $\mu_{\text{rad}}^2 = 0.670, 0.771, 1.190, 1.093, 1.348$); asymptotic log-slopes -1.006 and $+1.23$ – $+1.27$; mass-independent floor 27.03
 304 steps at $\gamma = 0.05$.*

Table 1: The $T > 0$ face of the budget, all four entries measured on the designed testbed.

stored direction	$\partial n_{1/2}/\partial \gamma$ (more friction)	$\partial n_{1/2}/\partial T$ (more heat)
coset / flat (designed)	> 0 , friction <i>preserves</i> ($+0.955 \pm 0.042$; $3.77 \times$ for $\gamma : 0.05 \rightarrow 0.2$)	< 0 , temperature <i>erases</i> (-0.956 to -0.979)
massive (underdamped)	< 0 , friction shortens (slope -1.006)	≈ 0 ; T sets only the fluctuation floor (max/min $\leq 1.013, 9/9$)

305 B The emergent arm, and the V-curve on a second instrument

306 *Emergent MLP checkpoints, 3 seeds, plus a matched designed control $\Rightarrow$ evidence. Pre-registered; predictions filed before any harness ran.*

307 *Four instruments, identical fit windows ($\gamma < \gamma_{\text{crit}}/2.5$, $\gamma > 2.5\gamma_{\text{crit}}$, keyed to the Jacobian’s γ_{crit} so the windows are not a free parameter):*
 308 *I-J*, the one-step Jacobian, $n_{1/2} = \ln 2/(-\ln |\lambda_{\text{ret}}|)$; *I-R1*, rollout first crossing of $|\theta| \leq \delta/2$; *I-R2*, rollout last crossing; *I-R3* (primary),
 309 rollout envelope-rate fit over the late window, $R^2 \geq 0.9955$ (median 0.9996) on every cell.

310 *Finite write amplitude.* At $\delta = 0.05/0.2/0.5$ rad the I-R3 argmin is 0.8928/0.9007/0.8968 (seed 42), 0.9047/0.9045/0.9174 (43),
 311 0.9028/0.9037/0.9070 (44) — at most $+1.4\%$ over a $10 \times$ range — and the slopes move by at most 1.9% ; every $\delta = 0.5$ write relaxed
 312 all the way home ($|\theta_{\text{final}}| \leq 6.5 \times 10^{-10}$), so no seed’s write cleared its washboard barrier. The pre-registered early-window clause *fails*,

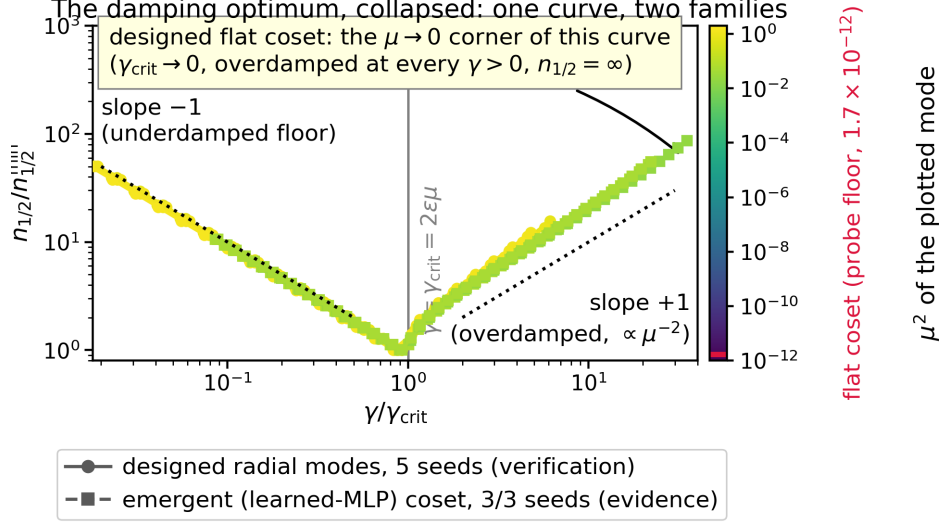


Figure 2: *The damping optimum, collapsed — the full-size version of Figure 1.* The same eight measured curves and the same ∓ 1 asymptotes, printed large enough to carry the annotations the main-text box cannot hold: the two-family legend, the $\gamma = \gamma_{\text{crit}} = 2\varepsilon\mu$ marker, the $\mu \rightarrow 0$ flat-coset callout, and the crimson probe-floor tick on the μ^2 colourbar (1.7×10^{-12}). No plotted value differs from Figure 1. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; langevin_noise="fat", Newtonian kinetic mode.*)

Table 2: The shape claims on all four instruments ($\delta = 0.05$ rad, 48 γ values, 3 emergent seeds). The rollout rate instrument reproduces the Jacobian; both threshold instruments fail below γ_{crit} , so no rollout $n_{1/2}$ is ever quoted from a threshold there.

	γ_{crit}	argmin/ γ_{crit} (I-J / I-R1 / I-R2 / I-R3)	slope below
seed 42	0.02334	0.8994 / 0.0857 / 0.2068 / 0.8928	-1.0023 / +0.0725 / -1.0556 / -0.9651
seed 43	0.01424	0.9046 / 0.1404 / 0.1404 / 0.9047	-1.0016 / +0.0688 / +0.0688 / -0.9977
seed 44	0.02265	0.9055 / 0.0883 / 0.3031 / 0.9028	-1.0022 / +0.0455 / -0.8854 / -0.9854
mean $\pm$ sd	0.9032 $\pm$ 0.0027 / 0.1048 $\pm$ 0.0252 / 0.2168 $\pm$ 0.0668 / 0.9001 $\pm$ 0.0052		-1.0020 $\pm$ 0.0003 / +0.0623 $\pm$ 0.0120 / -0.6240 $\pm$ 0.4948 / -0.9

Table 3: The instrument gap is a level, not a rate: on the overdamped branch ($\gamma \geq 2\gamma_{\text{crit}}$) the decay rates agree while the threshold instrument is offset by a per-checkpoint constant flat in γ . Seed 44's full-window value (0.9216 ± 0.0027) sits 0.0004 below its pre-registered band because that window still contains a second, faster mode (early-window ratio 0.539); the window-split diagnostic converges monotonically to 1 into the asymptote. Cause of the threshold offset, measured: the ring write deposits only a fraction of its amplitude in the slow branch (envelope-fit intercept over $\delta = 0.638/0.644/0.327$).

seed	$\Gamma_{\text{jac}}/\Gamma_{\text{R3}}$ (asymptotic window)	$n_{\text{jac}}/n_{\text{R1}}$ (threshold)	CV of $n_{\text{jac}}/n_{\text{R1}}$ over the grid
42	0.9935	1.3307	2.2–3.7% over 21–25 γ -points
43	1.0000	1.7981	
44	0.9948	1.8204	
mean	0.995 $\pm$ 0.003	per seed, at $\delta = 0.05$ rad	$ d \log(\text{ratio})/d \log \gamma \leq 0.0034$ (threshold 0.10)

313 $\Gamma_{\text{jac}}/\Gamma_{\text{roll}}^{\text{early}} = 0.52\text{--}0.85$ against a band of $[0.65, 1.35]$; the amplitude sweep identifies that as mode mixing rather than anharmonicity,
314 seed 44's early ratio being 0.539/0.527/0.522 across the same $10\times$ range.

315 *What the left branch does not contain.* For $\gamma < \gamma_{\text{crit}}$ the coset eigenpair is complex with $|\lambda|^2 = 1 - \gamma$, so $n_{1/2} = \ln 2 / (-\frac{1}{2} \ln(1 - \gamma))$
316 *independently of μ^2 and of the checkpoint:* at $\gamma = 0.002$, emergent seeds 42 ($\mu^2 = 5.449 \times 10^{-2}$) and 43 ($\mu^2 = 2.029 \times 10^{-2}$) both
317 print $\lambda_{\text{ret}} = 0.998999499$ and $n_{1/2} = 692.5$, against $2 \ln 2 / 0.002 = 693.1$. All model-dependent content of the V-curve is the argmin
318 location $\gamma_{\text{crit}} = 2\varepsilon\mu$ and the +1 branch; what the rollout adds is that the *nonlinear* model obeys that identity, which was not guaranteed.

319 *Designed control, and the model-side bound.* Over 5 designed seeds $\times$ 48 $\gamma \times$ 3 write amplitudes $\times$ 32000 steps of direct $T = 0$
320 rollout, $\max_n |\theta(n) - \delta| = 9.400 \times 10^{-13}$ rad, so $\Gamma \leq 7.309 \times 10^{-17}$ per step and $n_{1/2} \geq 9.483 \times 10^{15}$ steps against a
321 pre-registered bound of 10^{-6} rad; the matched Hessian μ_{soft}^2 there is 1.60×10^{-15} , 1.52×10^{-15} , -2.92×10^{-16} , 2.94×10^{-16} ,
322 1.21×10^{-16} . *Collapse:* in $x = \gamma/\gamma_{\text{crit}}$, $y = n_{1/2}\gamma_{\text{crit}}$ the three emergent curves agree to a median 1.15% on I-J and 5.67% on
323 I-R3; $y_{\text{min}} = 1.579/1.527/1.499$ and $1.631/1.535/1.662$ sit 22–24% below the continuum $2\sqrt{2} \ln 2 = 1.961$ and $x_{\text{min}} \approx 0.90$

sits 27% above the continuum 0.707, both known discrete-map corrections (matched designed control $x_{\min} = 0.83\text{--}0.93$). *Scope:* μ_{soft}^2 spans only $2.029 \times 10^{-2}\text{--}5.449 \times 10^{-2}$ across the three seeds, a factor 2.7, so the eleven-decade span is one decade of emergent variation attached to the designed $\mu \rightarrow 0$ corner and the rollout does not widen it.

The emergent arm's own negative. $1 - |\lambda_{\text{coset}}| = 1.06 \times 10^{-3}\text{--}2.96 \times 10^{-3}$ at $\gamma \approx 0.048$ against a designed $\leq 1.1 \times 10^{-15}$, with a whole-grid emergent minimum of $7.6 \times 10^{-5}\text{--}2.0 \times 10^{-4}$ against a designed grid minimum of exactly 0. Written $\delta \in \{0.1, 0.3, 0.5\}$ rad relaxes completely ($|\text{retained}| \leq 2.1 \times 10^{-3}$) where the designed checkpoint retains δ exactly; capacity is $\approx 1\text{--}1.6$ bits, not a continuum. Bias-corrected $n_{1/2}^{\text{obs}}/n_{1/2}^{\text{Goldstone}}$ at $\gamma = 0.2$, $\ell_{\theta}/\Delta < 0.06$, runs 2.75, 3.40 at $T = 2 \times 10^{-3}$ to 0.94–1.25 at $T \geq 4 \times 10^{-3}$, giving $T^* \approx 3 \times 10^{-3}$ (predicted $2.72\text{--}3.66 \times 10^{-3}$) against a washboard barrier of $2.29\text{--}3.57 \times 10^{-2}$. *Reporting rule:* raw exponents do not discriminate — a matched designed control gives $d \log n_{1/2}/d \log T = -0.53, -0.60, -1.04$ and $d \log n_{1/2}/d \log \gamma = +0.78, +0.63, +0.55$, indistinguishable from emergent, because both families carry the ℓ_{θ}/Δ boundary layer (up to 2.03). Only $|\lambda_{\text{coset}}|$ and bias-corrected deep-diffusive cells discriminate.

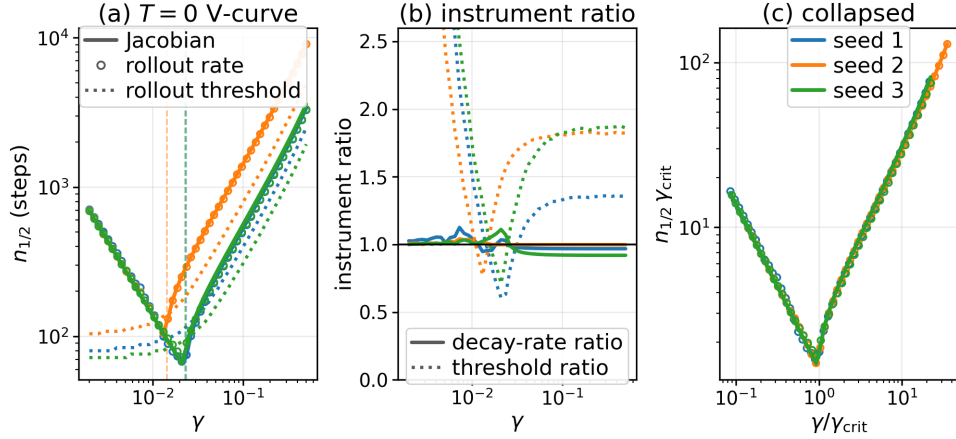


Figure 3: The V-curve measured a second way. (a) $T = 0$ $n_{1/2}(\gamma)$ per emergent seed on the one-step Jacobian (I-J, lines), the rollout envelope-rate instrument (I-R3, circles) and the rollout first-crossing threshold (I-R1, dotted), at $\delta = 0.05$ rad. (b) The per- γ instrument ratio: the rate ratio sits on 1, the threshold ratio is offset and flat in γ above γ_{crit} . (c) The same data in collapsed variables. (*Emergent MLP, 3 seeds, dim 4, hidden 64, $\varepsilon = 0.05$, float64, $T = 0$, pre-registered $\Rightarrow$ evidence.*)

335 C The friction-hole vault, on both arms

336 *Designed $SO(2)$ plus an analytic field, 3 seeds $\Rightarrow$ verification; emergent MLP, 3 seeds $\Rightarrow$ evidence. Pre-registered on both arms. All cells*
 337 *langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad.*

338 *Mechanism.* The shipped field is absorb-only, $p \leftarrow (1 - \gamma_{\phi})(1 - \gamma)p + \sigma(\gamma)\xi$ with $\sigma^2 = MT\gamma(2 - \gamma)$ set by the scalar γ , so with
 339 $\gamma_{\text{eff}} = 1 - (1 - \gamma)(1 - \gamma_{\phi})$, $T_{\text{local}} = T\gamma(2 - \gamma)/[\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})]$ and $D_{\theta} = \varepsilon T\gamma(2 - \gamma)/(2F^2\gamma_{\text{eff}}^2)$: a vault of $(\gamma_{\text{eff}}/\gamma)^2$. A
 340 coupled bath would give $D_{\theta} \propto \gamma_{\text{eff}}^{-1}$ and a vault of $[\gamma_{\text{eff}}(2 - \gamma)]/[\gamma(2 - \gamma_{\text{eff}})] = 13.88\times$ at $\gamma = 0.05$, $\gamma_{\phi} = 0.5$. The hypotheses
 341 differ by 7.942 $\times$; the coupled-bath value is a refuted prediction and never the vault.

342 *Designed arm.* $\text{Var}(p_i)/(M_i T) = 0.9986$ ($\gamma_{\phi} = 0$), 0.36171 ± 0.00129 (0.1), 0.23044 (0.2), 0.17513 (0.3), 0.12600 ± 0.00031
 343 (0.5; absorb prediction 0.12591), so $T_{\text{local}} = 1.26 \times 10^{-4}$, a 7.94 $\times$ refrigerator. $\text{MSD}/\text{pred}_{\text{absorb}} = 1.0011 \pm 0.0215$ over 15 cells;
 344 the D_{θ} -based vault is 8.42 $\times$, 22.39 $\times$, 44.31 $\times$, 107.77 $\pm 4.78\times$ at $\gamma_{\phi} = 0.1/0.2/0.3/0.5$ (absorb prediction 110.25). The direct
 345 first-passage vault reads 86.97 $\pm 2.94\times$ and is boundary-layer biased on the outside arm ($\ell_{\theta}/\Delta = 0.079$), so 107.77 $\times$ is the quoted number
 346 and travels with its estimator's name. Scalar control at the same γ_{eff} : 13.28 $\pm 0.12\times$ with $\text{Var}(p) = M_i T$ intact, field/scalar 8.11 ± 0.37
 347 against a predicted 7.942. At $T = 0$ the hole erases nothing (drift $\leq 1.75 \times 10^{-12}$ rad per 200k steps inside and outside, 12/12 cells;
 348 $||\lambda|_{\text{max}} - 1| \leq 2.0 \times 10^{-15}$ over 27 cells including the hole edge) and attenuates the write instead: $\theta_{\text{settle}}(\text{in})/\theta_{\text{settle}}(\text{out}) = 0.09576$
 349 against the transport law's 0.095238.

350 *Emergent arm.* The coset there is a bounded mode ($\mu_{\text{adiab}}^2 = 2.77 \times 10^{-2}\text{--}8.93 \times 10^{-2}$), so the designed block estimator is biased low
 351 by up to 55% and is replaced by a window-free Ornstein-Uhlenbeck fit $\text{MSD}(n) = A(1 - e^{-n/\tau})$, $\hat{D} = A/(2\varepsilon\tau)$, cross-validated
 352 on designed before use (112.58 $\pm 1.09\times$ against the banked 107.77 $\pm 4.78\times$; 14.16 $\pm 1.38\times$ against 13.28 $\pm 0.12\times$; 8.03 ± 0.80
 353 against 8.11 ± 0.37).

354 *The diffusion law on the emergent arm.* On bounded cells, $\hat{D}_{\text{ou}}/D_{\text{absorb}} = 1.0480$ ($\gamma_{\phi} = 0$, 1 cell), 1.1026 ± 0.0799 (0.1, 4), $1.0158 \pm$
 355 0.0519 (0.2, 5), 1.0552 ± 0.0496 (0.3, 5), 1.0418 ± 0.0519 (0.5, 6); against the coupled bath the same cells give 0.1312 ± 0.0065 at
 356 $\gamma_{\phi} = 0.5$. Law-referenced vault 106.1 $\pm 5.0\times$ (6 cells; prediction 110.25 $\times$; designed 107.77 $\pm 4.78\times$). A measured/measured ratio
 357 over the same cells reads 297.8 $\pm 196.8\times$ (median 234.4 $\times$); it is recorded and is never the vault number, because 5 of the 6 $\gamma_{\phi} = 0$ cells
 358 fail the bounded test ($\sigma_{\theta}^{\text{ou}} = 0.42\text{--}885$ rad against a Boltzmann 0.24–0.65), while every $\gamma_{\phi} \geq 0.2$ cell passes.

359 *The contrast number is designed-only, and the failure mode is the control.* A scalar γ does not cool — $\text{Var}(p)/(MT) = 0.99288 \pm 0.02354$
 360 at $\gamma = 0.525$ with no field, where both hypotheses predict 1.0, against 0.126 for the field at the same γ_{eff} — so the mechanism statement (a
 361 friction field is not locally stronger friction; it is locally stronger friction *plus* a local refrigerator) stands on emergent. The contrast measures
 362 23.39 ± 10.06 (per-cell 9.14/25.75/26.96/21.70/41.31/15.48) against a pre-registered band [6.5, 9.5] and a falsifier at > 14 : it
 363 fired. The cause is the control arm, whose own $\hat{D}_{\text{ou}}/D_{\text{flat}}$ is 1.1768/3.7319/3.3782/2.7999/5.2284/2.0506 — at $\gamma = 0.525$ with
 364 no cooling the emergent coset is still anharmonically delocalised — while the field arm sits on the absorb law at 1.02–1.15. The *direction*
 365 transfers; the number 8.11 ± 0.37 is a designed quantity.

Table 4: The refrigerator law transfers exactly to an emergent register: $\text{Var}(p_i)/(M_i T)$ against the absorb-only prediction, 2048 walkers $\times$ 40 samples, 3 emergent seeds; over all 24 field cells observed/absorb = 0.9998 ± 0.0019 , and at $\gamma_\phi = 0.5$ $T_{\text{local}}/T = 0.12570$, a $7.955\times$ refrigerator against 7.942 predicted. The coupled bath predicts 1.0 everywhere and is measured at 0.2235.

T	γ_ϕ	γ_{eff}	absorb pred.	coupled pred.	measured	obs/absorb
4×10^{-3}	0.0	0.050	1.00000	1.0	0.99815 ± 0.00257	0.9981
4×10^{-3}	0.1	0.145	0.36249	1.0	0.36326 ± 0.00067	1.0021
4×10^{-3}	0.2	0.240	0.23082	1.0	0.23062 ± 0.00080	0.9991
4×10^{-3}	0.3	0.335	0.17480	1.0	0.17473 ± 0.00046	0.9996
4×10^{-3}	0.5	0.525	0.12591	1.0	0.12592 ± 0.00055	1.0001
8×10^{-3}	0.5	0.525	0.12591	1.0	0.12548 ± 0.00032	0.9966

Table 5: The hole confines an emergent register, and the laundering control shows friction alone does not. $T = 4 \times 10^{-3}$, 1024 walkers, burn-in 20 000–80 000 steps, spread by interquartile range; “hop” means $|\theta| > 1$ rad. The pre-registered stationary-spread comparison is *void* here and no in/out spread ratio may be quoted from these checkpoints: the control that must return exactly 1.000 returns 0.4586 ± 0.1181 .

seed	arm	γ_{eff}	σ_θ (IQR)	hop fraction	Boltzmann at T_{local}	obs/pred
42	no hole, $\gamma = 0.05$	0.050	0.40795	5.50%	0.24073	1.6946
42	scalar 0.525 (control)	0.525	0.20034	0.73%	0.24073	0.8322
42	γ_ϕ hole	0.525	0.06442	0.0000	0.08542	0.7542
43	no hole	0.050	1.17601	42.97%	0.46059	2.5533
43	scalar 0.525	0.525	0.35335	10.20%	0.46059	0.7672
43	γ_ϕ hole	0.525	0.08628	0.0000	0.16343	0.5279
44	no hole	0.050	0.36922	2.36%	0.24034	1.5363
44	scalar 0.525	0.525	0.21574	0.26%	0.24034	0.8977
44	γ_ϕ hole	0.525	0.07279	0.0000	0.08528	0.8535

366 *First passage, and the confound it exposed.* An emergent potential has no rotational symmetry, so a $\theta = \pi$ “outside” arm is not a vacuum:
367 $|\nabla V| = 0.0420/0.1049/0.0414$ there against 0.0000 at $\theta = 0$, $V(\pi) - V(0) = +0.03163/+0.06003/+0.04315$, and μ_{adiab}^2
368 differing by up to $2.15\times$ between the sites. Any emergent first-passage number quoted against a $\theta = \pi$ baseline is confounded and a same-site
369 baseline is required. Same-site vaults ($|\Delta\theta| \geq 0.5$ rad, $T = 4 \times 10^{-3}$, $\gamma = 0.05$, 256 walkers, hole radius 1.0): $> 1379\times$ (seed 42),
370 $35.5\times$ (43), $> 1290\times$ (44), the first and third being censoring-limited *lower bounds* (86.7% and 93.4% censored at a 10^6 -step cap; closing
371 them needs $\approx 10^7$ steps per walker). The pre-registered $> 300\times$ holds 2/3 and the falsifier fires on seed 43, already flagged as geometrically
372 atypical — softest μ^2 , shallowest barrier (6.8×10^{-3}), coset at $\sigma_\theta/\Delta = 0.92$ outside and 0.33 inside, never leaving the diffusive regime.
373 The confound’s signature is seed 44, whose $\theta = \pi$ first passage (15585) is $20\times$ its own $\theta = 0$ value (775) and whose scalar control reads
374 $0.27\times$: more friction apparently *speeding* escape, impossible for a stationary register.

375 *Scope carried by the whole emergent arm.* Three model seeds; a single PRNG stream per cell, so every spread is across model seeds, not noise
376 realizations; both temperatures above T^* while the designed study’s trustworthy-cell restriction is $T \leq 3 \times 10^{-3}$, so every measurement is
377 squeezed between T^* and delocalisation and only cells passing the bounded test support a law claim. The hole is a ball of radius 1.0 in the
378 full 4-D space, so excursions of order 1.0 ($\approx 4.5\sigma_r$) leave it — a plausible contributor to seed 43’s low vault, not separately measured. No
379 temperature field was built.

380 D Exact store-level deletion: prior art and the measured tables

381 *Designed atom store, dim 3, capacity 8–64, no learning anywhere $\Rightarrow$ verification of the mechanism’s exactness. Every deletion statement carries*
382 *its scope: store level only, under the three stated conditions, with recency-based eviction permanently excluded.*

383 *Prior art, in full.* Uniquely represented (canonical) data structures date to Snyder (1977) and Andersson & Ottmann (1995); *obliviousness*, a
384 distributional condition on the memory representation which explicitly does not require canonical representation, is Micciancio’s (1997); the
385 weak and strong history-independence notions used here are Naor & Teague’s (2001); the open-addressed realisation is Blleloch & Golovin’s
386 (2007), whose table is a stable matching between keys and slots under a global key priority, and it is that priority-greedy rule and that delete-time
387 fix-up cascade we use; Blleloch, Golovin & Vassilevska (2008) had already taken unique representation into computational geometry. We claim
388 no priority over order-independent placement and no novelty for the displacement rule or its delete-time repair. New is the composition: slots that
389 are a lattice packing, so the canonical representation carries a minimum-separation certificate; content that is a continuously decaying amplitude
390 whose decay commutes with deletion; a canonical object that is an energy function a dynamical read relaxes into; and a price that is negative
391 where strong history independence is known to cost as much as an exponential slowdown relative to the weak notion for comparison-based
392 structures (Buchbinder & Petrank, 2003). “Fix-up cascade” is our name for the NEXT-chain delete-time repair of their Figure 1; the algorithm is
393 theirs either way.

394 *Exactness, measured at zero tolerance.* 24 exhaustive orders at $n = 4$, 200 random orders at $n \in \{8, 16, 40, 64\}$, 100 write/delete
395 interleavings and mid-decay deletes: all bit-identical, minimum spacing exactly 1.540000; on the shipped store the centres, payloads,
396 amplitudes and active mask are byte-equal and $V(q)$ is byte-equal on 64 random queries. *The packing price is negative:* at $K = 64$ the
397 canonical rule admits 61/64 offered keys deterministically ($\sigma = 0$), per-offered 0.9531, rising to 64/64 when the sizing is inflated by $\times 1.05$,
398 where the stochastic refuse-and-relocate allocator admits 43/64 on the same seeds (per-offered 0.6719); delete-time churn is 2.836 moves per
399 delete at full load. *At overflow, exactness fails:* with 7 cells and 8 offers, $\text{AUC}(n_{\text{live}}) = 1.00000 \pm 0.0000$, white-box
400 address-depth $\text{AUC} = 0.91409 \pm 0.0813$, byte-equal fraction 0.0000. A pre-registration that failed informatively: $\text{AUC}(z_{\text{hole}}) \geq 0.85$
401 was registered there and measured 0.576, because the fix-up cascade *fills* the freed cell with a re-placed survivor.

402 *What decays and what does not.* Against an exact adversary the white-box per-example AUC is 1.000 at all 18 amplitude levels down to
403 the floor; what decays is the effect size, the depth gap being $0.3935 \cdot A$ to four decimals at every level (the pre-registered closed form
404 $1 - e^{-1/2}$) and d' falling $1\ 560 \rightarrow 79.6$, exactly linear in A — the pre-registered falsifier fired on the AUC metric, as registered. Against a

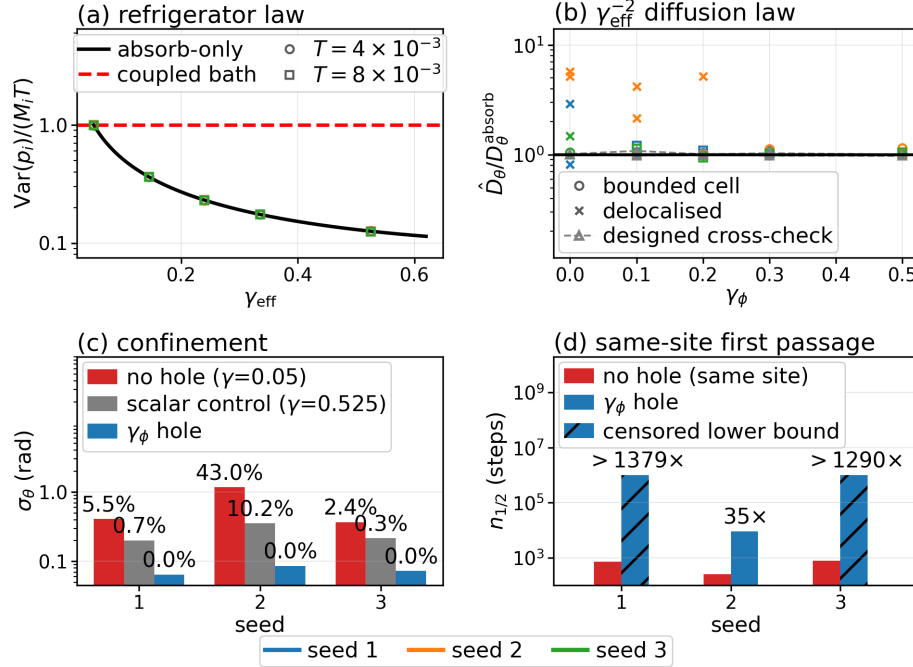


Figure 4: The vault on an emergent register; panels are labelled by the pre-registered prediction each tests. (a) The refrigerator law, $\text{Var}(p_i)/(M_i T)$ against γ_{eff} for 3 seeds $\times$ 2 temperatures, on the absorb-only curve with the coupled-bath prediction (dashed) rejected. (b) The γ_{eff}^{-2} diffusion law: bounded cells as circles and squares, delocalised cells as crosses, the designed cross-check overlaid. (c) Confinement: stationary spread and hop fraction for no hole, the scalar control and the hole. (d) Same-site first passage; hatched bars are censored lower bounds. (*Emergent MLP, 3 seeds, $T \in \{4, 8\} \times 10^{-3}$, both above T^* , $\Delta = 0.5$ rad $\Rightarrow$ evidence.*)

Table 6: The laundering control that fires. Designed store, shipped two-phase read, no learning anywhere; 8 targets $\times$ 3 seeds, 128 paired worlds per target, 18 amplitude levels. Against an *exact* adversary a boolean TTL flag inside the same store is as detectable as an $e^{-\lambda t}$ amplitude, so there is no measurable differentiator at the AUC level; the control fails to fire only against a resolution-limited adversary, and that resolution is our own modelling choice — which is why the main text leads with the retrieval geometry instead.

adversary	physical decay	TTL flag in the same store	separation
white-box, exact	1.000	1.000	0.000
query-level membership, exact	0.983	1.000	0.017
query-level membership, $\sigma_{\text{obs}} = 0.1$	0.559	0.996	0.437

405 resolution-limited adversary the grading is real and the pre-registered law survives, $A_{75} = 2.57\sigma_{\text{obs}}$ measuring 0.0780/0.2627/0.7537 at
 406 $\sigma_{\text{obs}} = 0.03/0.1/0.3$ against predicted 0.0771/0.2570/0.7710; but σ_{obs} is our modelling choice, so its defensible use is mechanistic,
 407 never protective. *State the $|c|$ distribution with any retention number:* every retention magnitude here is for the controller-placed disk (proposal
 408 radius 2.2869, packing bound 8.00); on a max-radius ring the same store gives retention 0.500 at $A = 0.06$ where this placement gives
 409 0.886. *Retrieval geometry:* R_{50} contracts 1.146/1.083/0.979/0.874/0.752 as $A : 1 \rightarrow 0.5 \rightarrow 0.2 \rightarrow 0.1 \rightarrow 0.06$, matching the
 410 pre-registered saddle prediction at all five points (± 0.20), where a TTL vector store's lookup radius is a constant hard step at 0.75–0.77.

411 *The trilemma, and what it costs.* The shipped store drops amplitude-independent address hold, which is why effective lifetime correlates
 412 $r = -0.85$ with a_i^2 ; the recommended gated-stiffness channel drops amplitude-independent latency and is the compute-adaptive-read dial
 413 — and it is *not* the shipped arm. Both previously proposed repairs are refuted: multiplying payloads by amplitudes kills the value criterion
 414 at $A = 1 - \text{tol}/|a_i|$ (measured death amplitudes 0.90/0.80/0.70/0.30 against the formula's 0.900/0.860/0.767/0.300) and inverts
 415 the payload dependence; launching the payload coordinate at the address-mapped point breaks the anti-decoration guard, after which a trivial
 416 substitute passes 0.672 of queries with no dynamics at all. Behind its flag the gated channel contracts R_{50} 1.135 $\rightarrow$ 0.771, retention std
 417 0.274 $\rightarrow$ 0.016, worst codeword 0.398 $\rightarrow$ 0.972; the registered payload-independence test *fails as a correlation* ($|r|$ only 0.846 $\rightarrow$ 0.667)
 418 and passes as an effect size; and at $\gamma_{\text{read}} = 0$ it is oscillatory, period $2\pi/\sqrt{\kappa(g_0 + A)}$ (log-log slope -0.5264), so no settling law may
 419 be quoted without a nonzero read friction. *No deletion-cost statement is made:* the shipped canonical synchronisation is an $O(n)$ rebuild per
 420 operation and the amortized-cost experiment has not been run.

421 E Prominent negatives

422 *Every negative below is on the record with its measurement; none is dropped, and each anchors a future-work item.*

423

claim tested	result
A continuous coset register emerges from training	No: an MLP potential on symmetric data yields a mid-spectrum massive mode ($1.7\text{--}4.9\times$ softer than the stiffest), capacity $\approx 1\text{--}1.6$ bits, complete relaxation of any written value. The $T = 0$ latch is designed-only.
Raw T/γ -exponents discriminate designed from emergent	No: a matched designed control reproduces the same shallow slopes; only ℓ_θ/Δ -controlled cells and $ \lambda_{\text{coset}} $ discriminate. Never an $n_{1/2}$ without Δ and ℓ_θ/Δ .
The friction hole thermalizes locally (coupled bath)	Rejected by a pre-registered measurement, by a factor 8.11: the field is absorb-only and the vault is $107.77 \pm 4.78\times$, against $13.28 \pm 0.12\times$ for the scalar control.
A friction field can <i>forget</i> a coset	No: at $T > 0$ it <i>lengthens</i> coset memory. Erasing latched content needs a temperature or tilt field — which explains the observed failure of a learned friction field to recover fit on a stationary vacuum.
Our own sign-flip prediction, in both observables	Falsified in one: a register written at the washboard minimum cannot flip back, and both signs coexist on one checkpoint for different observables. A sign-flip figure that does not name the initial condition is meaningless.
Amplitude decay reduces an exact adversary's distinguishability	No, at any point in an item's life: white-box AUC 1.000 at all 18 amplitude levels; what decays is the effect size. The graded curve exists only relative to a stated adversary resolution, our own modelling choice.
Canonical placement is exact at overflow without the waitlist	No: $\text{AUC}(n_{\text{live}}) = 1.000$, byte-equal fraction 0.0000. Every deletion sentence carries its scope clause; recency eviction is permanently excluded.
The placement algorithm is ours	No: Blleloch & Golovin (2007) own it outright, and unique representation in a geometric setting is also taken (2008). Our claim is the composition only.
A per-deletion cost claim can be made	No: the shipped synchronisation is an $O(n)$ rebuild per operation, so the amortized-cost formalism cannot be instantiated on our own code until that experiment runs.
The payload-lifetime coupling has a repair	Both proposed repairs are refuted, and the registered payload-independence statistic fails as a correlation while passing as an effect size. At zero read friction the gated channel is oscillatory, not relaxational.

425

claim tested	result
A pseudo-Goldstone tilt is a lifetime dial on a learned store	Refuted <i>in sign</i> : a tilt monotonically <i>reduces</i> the smallest eigenvalue on two independent implementations and every family, because a written site is not a common vacuum of its own atoms. The relation survives only in the single-atom geometry it was specified in.
A tilt sets the soft scale on a learned store	No: the confinement floors the soft eigenvalue at 2α , capping lifetime at $\tau_{\text{max}} = \Gamma/2\alpha$. The registered damped-mode floor is confirmed, the $1/\text{tilt}$ branch unreachable, and lowering the confinement breaks the write.
The lifecycle's protected-fraction leg is exercised	No: 0 refusals on the stream at the measured operating point. Reported as <i>unexercised</i> — not working and not broken.
A task-level payoff is claimed	No: forgetting here is shown predictable, controllable, removable and schedulable, not shown to improve a downstream metric. External benchmarks won on their own headline metric = ZERO.
The field/scalar vault contrast transfers to an emergent register	No: 23.39 ± 10.06 against 8.03 ± 0.80 from the identical estimator on designed; the pre-registered falsifier fired. The cause is the laundering control itself, off its own law by $1.18\text{--}5.23\times$. The number is designed-only; its direction transfers.
The emergent stationary-spread diagnostic is measurable above T^*	No: its own control, which must return 1.000, returns 0.4586 ± 0.1181 — the no-hole arm is a hopping rotor, so no in/out spread ratio may be quoted. The confinement statement replaces it.
A $\theta = \pi$ baseline is valid for emergent first passage	No: $ \nabla V = 0.042\text{--}0.105$ there against 0.0000 at $\theta = 0$, and one seed's scalar control reads $0.27\times$, i.e. more friction apparently speeding escape. Same-site baselines are required; the designed arm is protected by symmetry but was never checked, and that check is owed.
Threshold instruments are usable below γ_{crit}	No: a first-crossing $n_{1/2}$ puts its argmin at the grid edge with a below-slope of $+0.0623 \pm 0.0120$ where the law says -1 , and last-crossing carries an additive half-period bias. Only the envelope-rate instrument is quotable there.
Our microscopic explanation of the instrument offset	Wrong, though the offset itself is real, constant and quotable: the pre-registered single-slow-mode projection fails (amplitude fraction $0.638/0.644/0.327$ does not order like the angular overlaps) and the early-window ratio $0.52\text{--}0.85$ is amplitude-independent — multi-mode transient, not anharmonicity.
The emergent first-passage vault is measured	Only bounded: $> 1379\times$ and $> 1290\times$ at $86.7\%/93.4\%$ censoring, and $35.5\times$ on the geometrically atypical seed. Three seeds cannot say which is typical.

427 *Anonymization note.* Any supplementary or linked material, including code, is anonymized; only an anonymized snapshot may be linked.